

All Formulas From Chatnotes

This sheet consolidates the formulas, model equations, hypothesis tests, and exam templates contained in the **chatnotes** folder. Repeated worked-example substitutions are consolidated into their underlying formula so the sheet is usable for PDF export.

Conventions

$$t, T, h, k, n, p, q, N$$

Variables: t is a time index; T is sample size or the final observed time; h is a forecast horizon; k is a lag or number of model parameters depending on context; n is a return horizon; p, q are lag orders; N is the number of assets.

$$F_t$$

Variables: F_t is the information set available at time t .

Returns may be written as decimals or percentages. Check the scale before applying VaR or standard deviation formulas.

Prices, Returns, and VaR

Simple return

$$R_t = \frac{P_t - P_{t-1}}{P_{t-1}} = \frac{P_t}{P_{t-1}} - 1$$

Variables: R_t is the simple return from $t-1$ to t ; P_t is the price at time t ; P_{t-1} is the previous price.

Gross return

$$1 + R_t = \frac{P_t}{P_{t-1}}$$

Variables: $1 + R_t$ is the gross return, the multiplier applied to wealth.

Return with dividends

$$R_t = \frac{P_t + D_t - P_{t-1}}{P_{t-1}}$$

Variables: D_t is the dividend paid during period t .

Log return

$$r_t = \log(1 + R_t) = \log(P_t) - \log(P_{t-1}) = \log\left(\frac{P_t}{P_{t-1}}\right)$$

Variables: r_t is the log return; $\log$ is the natural logarithm.

Small-return approximation

$$\log(1 + R_t) \approx R_t$$

Variables: R_t is the simple return; the approximation works best when R_t is small.

Multi-period gross return

$$1 + R_t(k) = \prod_{j=0}^{k-1} (1 + R_{t-j})$$

Variables: $R_t(k)$ is the k -period simple return ending at t ; j indexes the returns being multiplied.

Multi-period simple return

$$R_t(k) = \prod_{j=0}^{k-1} (1 + R_{t-j}) - 1$$

Variables: $R_t(k)$ is the total simple return over k periods.

Equity curve for one dollar

$$V_t(k) = \prod_{j=0}^{k-1} (1 + R_{t-j})$$

Variables: $V_t(k)$ is the value at time t of one dollar invested over k periods.

Wealth growth

$$V_T = V_0 \prod_{t=1}^T (1 + R_t)$$

Variables: V_0 is starting wealth; V_T is ending wealth after T periods.

Multi-period log return

$$r_t(k) = \sum_{j=0}^{k-1} r_{t-j} = r_t + r_{t-1} + \dots + r_{t-k+1}$$

Variables: $r_t(k)$ is the k -period log return.

Total log return between two prices

$$r(1, T) = \sum_{t=1}^T r_t = \log\left(\frac{P_T}{P_0}\right)$$

Variables: P_0 is the starting price; P_T is the ending price.

Index log return

$$r_t = \log(Index_t) - \log(Index_{t-1})$$

Variables: $Index_t$ is the index level at time t .

Excess returns

$$\begin{aligned}\text{asset excess return} &= \text{asset return} - R_f \\ \text{market excess return} &= \text{market return} - R_f\end{aligned}$$

Variables: R_f is the risk-free return.

Market beta interpretation

$$\begin{aligned}\beta > 1 &\Rightarrow \text{aggressive, more sensitive than the market} \\ \beta = 1 &\Rightarrow \text{tracks the market} \\ 0 < \beta < 1 &\Rightarrow \text{defensive, less sensitive than the market} \\ \beta = 0 &\Rightarrow \text{unrelated to market movements}\end{aligned}$$

Variables: β is market sensitivity.

Market capitalisation

market capitalisation = share price $\times$ number of shares outstanding

Variables: market capitalisation is firm size measured using market value.

Book-to-market ratio

$$\text{BtM} = \frac{\text{book value of equity}}{\text{market value of equity}}$$

Variables: BtM is the book-to-market ratio.

Normal return model

$$R \sim \mathcal{N}(\mu, \sigma^2)$$

Variables: R is a return; μ is mean return; σ^2 is variance; σ is standard deviation.

Normal quantile

$$q_\alpha = \mu + \sigma z_\alpha$$

Variables: q_α is the lower-tail return quantile; α is the tail probability; z_α is the standard normal cutoff.

Common normal cutoffs

$$z_{0.05} = -1.645$$

$$z_{0.025} = -1.96$$

Variables: $z_{0.05}$ is the 5 percent lower-tail cutoff; $z_{0.025}$ is the 2.5 percent lower-tail cutoff.

One-asset VaR

$$\text{VaR}_\alpha = |W_0 q_\alpha|$$

Variables: VaR_α is Value at Risk at tail probability α ; W_0 is the dollar value invested.

Zero-mean 5 percent normal VaR

$$q_{0.05} = -1.645\sigma$$

$$\text{VaR}_{0.05} = |W_0(-1.645\sigma)|$$

Variables: $q_{0.05}$ is the 5 percent return cutoff; σ is return volatility.

Random Variables and Distribution Properties

Probability under a density

$$\Pr(a \leq X \leq b) = \text{area under } f(x) \text{ from } a \text{ to } b$$

Variables: X is a random variable; a, b are cutoffs; $f(x)$ is the probability density.

Normal distribution notation

$$X \sim \mathcal{N}(\mu, \sigma^2)$$

Variables: X is a random variable; μ is the mean; σ^2 is the variance.

Expected value

$$\mathbb{E}[X] = \mu$$

Variables: $\mathbb{E}[\cdot]$ means expected value.

State-contingent expected return

$$\mathbb{E}(R) = \sum_s R_s \Pr(S = s)$$

Variables: R_s is the return in state s ; S is the state variable.

Expected value linearity

$$\begin{aligned}\mathbb{E}(a) &= a \\ \mathbb{E}(aX) &= a\mathbb{E}(X) \\ \mathbb{E}(a + X) &= a + \mathbb{E}(X) \\ \mathbb{E}(X + Y) &= \mathbb{E}(X) + \mathbb{E}(Y)\end{aligned}$$

Variables: a is a constant; X, Y are random variables.

Variance

$$\text{Var}(X) = \mathbb{E}[(X - \mu)^2]$$

Variables: $\text{Var}(X)$ is the variance of X .

Standard deviation

$$\text{sd}(X) = \sqrt{\text{Var}(X)}$$

Variables: $\text{sd}(X)$ is the standard deviation of X .

Variance rules

$$\begin{aligned}\text{Var}(a) &= 0 \\ \text{Var}(aX) &= a^2 \text{Var}(X) \\ \text{Var}(a + X) &= \text{Var}(X)\end{aligned}$$

Variables: a is a constant.

Variance of a sum

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$$

Variables: $\text{Cov}(X, Y)$ is covariance between X and Y .

Standardising a normal variable

$$Z = \frac{X - \mu}{\sigma} \sim \mathcal{N}(0, 1)$$

Variables: Z is a standard normal variable; σ is the standard deviation of X .

Probability conversion after standardising

$$\Pr(X > a) = \Pr\left(\frac{X - \mu}{\sigma} > \frac{a - \mu}{\sigma}\right) = \Pr(Z > z)$$

Variables: $z = (a - \mu)/\sigma$ is the standardised cutoff.

Lower-tail normal probability

$$\Pr(r_t < x) = \Pr\left(Z < \frac{x - \mu}{\sigma}\right)$$

Variables: r_t is a return; x is a cutoff.

Return scale convention

0.01 = 1 percent, if returns are decimals

1 = 1 percent, if returns are stored as percentage points

Variables: the same economic return can be represented on different numeric scales.

Sample mean

$$\bar{r} = \frac{1}{T} \sum_{t=1}^T r_t$$

Variables: $\bar{r}$ is the sample average return.

Sample variance and standard deviation

$$s^2 = \frac{1}{T-1} \sum_{t=1}^T (r_t - \bar{r})^2, \quad s = \sqrt{s^2}$$

Variables: s^2 is sample variance; s is sample standard deviation.

Skewness and kurtosis

$$\text{skewness} = \frac{\mathbb{E}[(r_t - \mu)^3]}{\sigma^3}, \quad \text{kurtosis} = \frac{\mathbb{E}[(r_t - \mu)^4]}{\sigma^4}$$

Variables: skewness measures asymmetry; kurtosis measures tail thickness.

Co-Movement, Conditional Moments, and Factors

Joint probability

$$\Pr(X = x, Y = y)$$

Variables: X, Y are random variables; x, y are outcomes.

Marginal probability

$$\Pr(X = x) = \sum_y \Pr(X = x, Y = y)$$

Variables: the sum is over all possible values of Y .

Conditional probability

$$\Pr(X = x \mid Y = y) = \frac{\Pr(X = x, Y = y)}{\Pr(Y = y)}$$

Variables: $\Pr(X = x \mid Y = y)$ is the probability of $X = x$ given $Y = y$.

Independence

$$\Pr(X = x \mid Y = y) = \Pr(X = x)$$

Variables: independence means knowing Y does not change the probability distribution of X .

Conditional expectation

$$\mathbb{E}[X | Y]$$

Variables: $\mathbb{E}[X | Y]$ is the expected value of X conditional on Y .

Law of iterated expectations

$$\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X | Y]]$$

Variables: Y is the conditioning variable.

Conditional variance

$$\text{Var}(Y | X) = \mathbb{E}[(Y - \mathbb{E}[Y | X])^2 | X]$$

Variables: $\text{Var}(Y | X)$ is remaining uncertainty about Y after conditioning on X .

Law of total variance

$$\text{Var}(Y) = \mathbb{E}[\text{Var}(Y | X)] + \text{Var}(\mathbb{E}[Y | X])$$

Variables: the first term is average conditional variance; the second is variance of conditional means.

Information reduces conditional variance

$$\text{Var}(Y | \text{information}) < \text{Var}(Y)$$

Variables: information is any conditioning set that improves prediction.

Covariance

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])]$$

Variables: covariance measures linear co-movement.

State covariance

$$\text{Cov}(R_A, R_B) = \sum_s p_s (R_{A,s} - \mu_A)(R_{B,s} - \mu_B)$$

Variables: R_A, R_B are asset returns; p_s is the probability of state s ; μ_A, μ_B are expected returns.

Covariance from correlation

$$\text{Cov}(R_A, R_B) = \rho_{AB} \sigma_A \sigma_B$$

Variables: ρ_{AB} is the correlation between assets A and B; σ_A, σ_B are standard deviations.

Correlation

$$\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\text{sd}(X) \text{sd}(Y)}$$

Variables: $\text{Corr}(X, Y)$ is the standardised covariance.

Correlation from covariance

$$\rho_{AB} = \frac{\text{Cov}(R_A, R_B)}{\sigma_A \sigma_B}$$

Variables: ρ_{AB} is the correlation coefficient.

Covariance sign interpretation

$$\begin{aligned}\text{Cov}(X, Y) > 0 &\Rightarrow X \text{ and } Y \text{ tend to move together} \\ \text{Cov}(X, Y) < 0 &\Rightarrow X \text{ and } Y \text{ tend to move oppositely} \\ \text{Cov}(X, Y) = 0 &\Rightarrow \text{no linear co-movement}\end{aligned}$$

Variables: the sign of covariance gives direction of linear co-movement.

Group mean return

$$\bar{r}_g = \frac{1}{T} \sum_{t=1}^T r_{g,t}$$

Variables: $\bar{r}_g$ is the mean return for group g ; $r_{g,t}$ is group return at time t .

Generic factor long-short return

$$r_{\text{factor},t} = r_{\text{long},t} - r_{\text{short},t}$$

Variables: $r_{\text{long},t}$ is the return on the long leg; $r_{\text{short},t}$ is the return on the short leg.

SMB factor

$$SMB_t = R_{\text{small},t} - R_{\text{big},t}$$

Variables: SMB_t is Small Minus Big at time t .

HML factor

$$HML_t = R_{\text{high BtM},t} - R_{\text{low BtM},t}$$

Variables: HML_t is High Minus Low book-to-market at time t .

Fama-French two-by-three SMB construction

$$SMB = \text{average}(S/L, S/N, S/H) - \text{average}(B/L, B/N, B/H)$$

Variables: S means small; B means big; L, N, H mean low, neutral, and high book-to-market portfolios.

Fama-French two-by-three HML construction

$$HML = \text{average}(S/H, B/H) - \text{average}(S/L, B/L)$$

Variables: S/H is the small high-BtM portfolio; B/H is the big high-BtM portfolio; $S/L, B/L$ are low-BtM portfolios.

Portfolio Risk and Sharpe Ratios

Two-asset portfolio return

$$R_p = wR_1 + (1 - w)R_2$$

Variables: R_p is portfolio return; w is the weight in asset 1; R_1, R_2 are asset returns.

Two-asset expected return

$$\mathbb{E}[R_p] = w\mathbb{E}[R_1] + (1 - w)\mathbb{E}[R_2]$$

Variables: $\mathbb{E}[R_p]$ is expected portfolio return.

Two-asset expected return using means

$$\mu_p = w\mu_1 + (1 - w)\mu_2$$

Variables: μ_p is portfolio mean return; μ_1, μ_2 are asset mean returns.

Two-asset portfolio variance

$$\text{Var}(R_p) = w^2\sigma_1^2 + (1 - w)^2\sigma_2^2 + 2w(1 - w)\sigma_{12}$$

Variables: σ_1^2, σ_2^2 are asset variances; σ_{12} is covariance.

Two-asset portfolio variance using explicit weights

$$\sigma_p^2 = w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\sigma_{12}$$

Variables: w_1, w_2 are portfolio weights; σ_p^2 is portfolio variance.

Covariance from correlation

$$\sigma_{12} = \rho_{12}\sigma_1\sigma_2$$

Variables: ρ_{12} is correlation between assets 1 and 2.

Sharpe ratio

$$SR = \frac{\text{mean return} - \text{risk-free rate}}{\text{standard deviation}}$$

Variables: SR is the Sharpe ratio.

Asset Sharpe ratio

$$SR_i = \frac{\mu_i - r_f}{\sigma_i}$$

Variables: μ_i is expected return on asset i ; r_f is the risk-free rate; σ_i is standard deviation.

Portfolio Sharpe ratio

$$SR_p = \frac{\mu_p - r_f}{\sigma_p}$$

Variables: SR_p is portfolio Sharpe ratio; σ_p is portfolio standard deviation.

Two-asset GMV weight

$$w_1^{GMV} = \frac{\sigma_2^2 - \sigma_{12}}{\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}}$$

Variables: w_1^{GMV} is the global minimum variance weight in asset 1.

Second GMV weight

$$w_2^{GMV} = 1 - w_1^{GMV}$$

Variables: w_2^{GMV} is the GMV weight in asset 2.

Regression trick for GMV weights

$$r_{2,t} = \beta_0 + \beta_1(r_{2,t} - r_{1,t}) + e_t$$

Variables: $r_{1,t}, r_{2,t}$ are asset returns; β_0, β_1 are regression coefficients; e_t is the error.

$$\beta_1 = w_1, \quad w_2 = 1 - \beta_1$$

Variables: β_1 gives the weight in asset 1 under the regression trick.

Regression slope form used in practice

$$b = \frac{\text{Cov}(R_1, R_2)}{\text{Var}(R_2)}$$

Variables: b is a regression slope in the practice note.

Many-asset GMV weights

$$\mathbf{w}^{GMV} = \frac{\Sigma^{-1}\mathbf{1}}{\mathbf{1}'\Sigma^{-1}\mathbf{1}}$$

Variables: $\mathbf{w}^{GMV}$ is the GMV weight vector; Σ is the covariance matrix; $\mathbf{1}$ is a vector of ones.

Portfolio return from GMV weights

$$r_{p,t} = w_1 r_{1,t} + (1 - w_1) r_{2,t}$$

Variables: $r_{p,t}$ is portfolio return at time t .

Asset Pricing Models

CAPM pricing equation

$$\mathbb{E}(R_i) - r_f = \beta_i [\mathbb{E}(R_m) - r_f]$$

Variables: R_i is asset return; R_m is market return; β_i is market beta.

CAPM regression

$$r_{i,t} - r_{f,t} = \alpha_i + \beta_i(r_{m,t} - r_{f,t}) + e_{i,t}$$

Variables: $r_{i,t} - r_{f,t}$ is asset excess return; $r_{m,t} - r_{f,t}$ is market excess return; α_i is abnormal return; $e_{i,t}$ is the residual.

CAPM alpha restriction

$$\alpha_i = 0$$

Variables: α_i is abnormal return relative to CAPM.

CAPM total risk decomposition

$$\text{Var}(r_{i,t} - r_f) = \beta_i^2 \text{Var}(r_{m,t} - r_f) + \text{Var}(e_{i,t})$$

Variables: total excess-return variance equals systematic variance plus idiosyncratic variance.

Systematic and idiosyncratic fractions

$$\begin{aligned} \text{systematic fraction} &= R^2 \\ \text{idiosyncratic fraction} &= 1 - R^2 \end{aligned}$$

Variables: R^2 is the regression explained-variation statistic.

Fama-French three-factor model

$$r_{i,t} - r_{f,t} = \alpha_i + \beta_{i1}(r_{m,t} - r_{f,t}) + \beta_{i2}SMB_t + \beta_{i3}HML_t + e_{i,t}$$

Variables: β_{i1} is market loading; β_{i2} is SMB loading; β_{i3} is HML loading.

Carhart four-factor model

$$r_{i,t} - r_{f,t} = \alpha_i + \beta_{i1}(r_{m,t} - r_{f,t}) + \beta_{i2}SMB_t + \beta_{i3}HML_t + \beta_{i4}MOM_t + e_{i,t}$$

Variables: MOM_t is the momentum factor; β_{i4} is momentum loading.

CAPM nested in the four-factor model

$$\beta_{i2} = \beta_{i3} = \beta_{i4} = 0$$

Variables: setting the extra factor loadings to zero reduces the four-factor model to CAPM.

Single-coefficient t-test

$$\begin{aligned} H_0 : \beta &= \beta_0 \\ H_1 : \beta &\neq \beta_0 \\ t &= \frac{\hat{\beta} - \beta_0}{\text{se}(\hat{\beta})} \end{aligned}$$

Variables: $\hat{\beta}$ is the estimated coefficient; β_0 is the hypothesised value; $\text{se}(\hat{\beta})$ is the standard error.

Test whether a factor matters

$$\begin{aligned} H_0 : \beta_{SMB} &= 0 \\ H_1 : \beta_{SMB} &\neq 0 \\ t &= \frac{\hat{\beta}_{SMB}}{\text{se}(\hat{\beta}_{SMB})} \end{aligned}$$

Variables: β_{SMB} is the SMB coefficient.

Joint test of CAPM against a multi-factor model

$$\begin{aligned} H_0 : \beta_{SMB} &= \beta_{HML} = \beta_{MOM} = 0 \\ H_1 : &\text{at least one extra factor coefficient is non-zero} \\ J &= \frac{RSS_0 - RSS_1}{RSS_1/(T - K - 1)} \end{aligned}$$

Variables: RSS_0 is restricted-model residual sum of squares; RSS_1 is unrestricted-model residual sum of squares; K is the number of regressors in the unrestricted model.

Residual standard error and RSS

$$RSS = (\text{residual standard error})^2 df$$

Variables: df is residual degrees of freedom.

$$\text{residual standard error} = \sqrt{\frac{RSS}{df}}$$

Variables: residual standard error estimates residual standard deviation.

$$df = T - K - 1$$

Variables: T is sample size; K is the number of regressors excluding the intercept.

Regression Diagnostics and EMH Tests

Regression residual properties

$$\begin{aligned}\mathbb{E}(e_{i,t}) &= 0 \\ \text{Var}(e_{i,t}) &\text{ is constant} \\ \mathbb{E}(e_{i,t}e_{i,t-k}) &= 0, \quad k = 1, 2, \dots\end{aligned}$$

Variables: $e_{i,t}$ is a regression disturbance or residual.

Original model and residual

$$y_t = \alpha + \beta x_t + e_t$$

Variables: y_t is the dependent variable; x_t is the regressor; α, β are coefficients.

$$\hat{e}_t = y_t - \hat{\alpha} - \hat{\beta}x_t$$

Variables: $\hat{e}_t$ is the fitted residual.

Heteroskedasticity hypotheses

$$\begin{aligned}H_0 : \text{Var}(e_{i,t}) &\text{ is constant} \\ H_1 : \text{Var}(e_{i,t}) &\text{ is not constant}\end{aligned}$$

Variables: the null is homoskedasticity; the alternative is heteroskedasticity.

Heteroskedasticity auxiliary regression

$$\hat{e}_t^2 = \gamma_0 + \gamma_1 x_t + \gamma_2 x_t^2 + v_t$$

Variables: γ_j are auxiliary-regression coefficients; v_t is the auxiliary error.

LM statistic for heteroskedasticity

$$W = TR^2$$

Variables: R^2 comes from the auxiliary regression; compare W with a chi-square critical value.

Residual autocorrelation hypotheses

$$\begin{aligned}H_0 : &\text{ no autocorrelation} \\ H_1 : &\text{ at least one lagged residual matters}\end{aligned}$$

Variables: the alternative means residual serial correlation remains.

Residual autocorrelation auxiliary regression

$$\hat{e}_t = \gamma_0 + \gamma_1 \hat{e}_{t-1} + \gamma_2 \hat{e}_{t-2} + \dots + v_t$$

Variables: $\hat{e}_{t-j}$ are lagged residuals.

LM statistic for autocorrelation

$$AR(p) = TR^2$$

Variables: p is the number of residual lags; under the null compare to χ_p^2 .

HAC definition

HAC = heteroskedasticity and autocorrelation consistent

Variables: HAC/Newey-West changes standard errors, not fitted coefficients.

ETF tracking regression

$$r_{\text{ETF},t} = \alpha + \beta r_{\text{index},t} + e_t$$

Variables: β measures the ETF's exposure to the index.

Leveraged ETF target test

$$\begin{aligned} H_0 : \beta &= -2 \\ H_1 : \beta &\neq -2 \\ t &= \frac{\hat{\beta} - (-2)}{\text{se}(\hat{\beta})} \end{aligned}$$

Variables: $\hat{\beta}$ is the estimated tracking slope.

EMH return model

$$r_t = \mu + e_t$$

Variables: μ is constant mean return; e_t is the unpredictable return component.

White noise notation

$$e_t \sim WN(0, \sigma^2)$$

Variables: $WN(0, \sigma^2)$ means white noise with zero mean and variance σ^2 .

White noise conditions

$$\begin{aligned} \mathbb{E}(e_t) &= 0 \\ \text{Var}(e_t) &= \sigma^2 \\ \text{Corr}(e_t, e_{t-k}) &= 0, \quad k \geq 1 \end{aligned}$$

Variables: no autocorrelation means past shocks do not linearly predict current shocks.

Autocorrelation

$$\rho(k) = \text{Corr}(r_t, r_{t-k})$$

Variables: $\rho(k)$ is autocorrelation at lag k .

Correlogram confidence band

$$\pm \frac{2}{\sqrt{T}}$$

Variables: T is sample size; the band is an approximate 5 percent individual significance guide.

Q-test hypotheses

$$\begin{aligned} H_0 : \rho_1 &= \rho_2 = \dots = \rho_k = 0 \\ H_1 : &\text{at least one autocorrelation is non-zero} \end{aligned}$$

Variables: ρ_j is autocorrelation at lag j .

Portmanteau/Ljung-Box statistic from the practice exam

$$Q = T(T+2) \sum_{k=1}^m \frac{\hat{\rho}_k^2}{T-k}$$

Variables: m is the largest lag tested; $\hat{\rho}_k$ is the sample autocorrelation at lag k ; compare Q with χ_m^2 .

Single autocorrelation large-sample test

$$\sqrt{T} \hat{\rho}(k) \approx \mathcal{N}(0, 1)$$

Variables: $\hat{\rho}(k)$ is the sample autocorrelation at lag k .

$$\begin{aligned} H_0 : \rho(k) &= 0 \\ H_1 : \rho(k) &\neq 0 \\ \left| \sqrt{T} \hat{\rho}(k) \right| &> 1.96 \Rightarrow \text{reject at approximately 5 percent} \end{aligned}$$

Variables: 1.96 is the two-sided 5 percent normal cutoff.

Demeaned returns for LM predictability tests

$$e_t = r_t - \bar{r}$$

Variables: $\bar{r}$ is the sample mean return.

LM return predictability regression

$$e_t = \gamma_0 + \gamma_1 e_{t-1} + \gamma_2 e_{t-2} + \dots + v_t$$

Variables: lagged demeaned returns test predictability.

LM return predictability hypotheses and statistic

$$\begin{aligned} H_0 : \gamma_1 &= \gamma_2 = \dots = \gamma_k = 0 \\ H_1 : &\text{at least one } \gamma_j \text{ differs from zero} \\ AR(k) &= TR^2 \sim \chi_k^2 \end{aligned}$$

Variables: γ_j are lag coefficients; χ_k^2 is chi-square with k degrees of freedom.

Variance-ratio one-period log return

$$r_t = \log(P_t) - \log(P_{t-1})$$

Variables: r_t is one-period log return.

Variance-ratio n-period log return

$$r_t(n) = r_t + r_{t-1} + \dots + r_{t-(n-1)}$$

Variables: $r_t(n)$ is the n -period log return.

Variance scaling under independence

$$\text{Var}[n\text{-period return}] = n \text{Var}[1\text{-period return}]$$

Variables: this holds approximately when returns are independent over time.

Variance ratio

$$VR_n = \frac{\text{variance of } n\text{-period returns}}{n \times \text{variance of 1-period returns}}$$

Variables: $VR_n = 1$ suggests no autocorrelation; $VR_n > 1$ suggests positive autocorrelation; $VR_n < 1$ suggests negative autocorrelation.

Stationarity, Random Walks, ADF, and AR Models

Covariance stationarity

$$\begin{aligned}\mathbb{E}(Y_t) &= \mu \\ \text{Var}(Y_t) &= \sigma^2 \\ \text{Cov}(Y_t, Y_{t-j}) &= \gamma_j\end{aligned}$$

Variables: stationarity means constant mean, constant variance, and autocovariance depending only on lag j .

Stationarity rule of thumb

$$\begin{array}{ll}\log \text{ prices} & \text{often non-stationary} \\ \log \text{ returns} & \text{often stationary}\end{array}$$

Variables: this is a modelling guide, not a formal test.

Random walk

$$p_t = p_{t-1} + \epsilon_t$$

Variables: p_t is usually log price; ϵ_t is the shock.

Random walk successive substitution

$$p_t = p_0 + \epsilon_1 + \dots + \epsilon_t$$

Variables: p_0 is the initial value.

Random walk moments

$$\mathbb{E}(p_t) = p_0, \quad \text{Var}(p_t) = \sigma^2 t$$

Variables: σ^2 is shock variance. If $p_0 = 0$, the note's simplified mean is $\mathbb{E}(p_t) = 0$.

Random walk with drift

$$p_t = \mu + p_{t-1} + \epsilon_t$$

Variables: μ is drift.

Random walk with drift successive substitution

$$p_t = p_0 + \mu t + \epsilon_1 + \dots + \epsilon_t$$

Variables: drift accumulates linearly over time.

Random walk with drift moments

$$\mathbb{E}(p_t) = p_0 + \mu t, \quad \text{Var}(p_t) = \sigma^2 t$$

Variables: if $p_0 = 0$, the note's simplified mean is μt .

Unit-root AR form

$$y_t = \phi_1 y_{t-1} + \text{error}_t$$

Variables: ϕ_1 is the autoregressive coefficient.

Unit root condition

$$\phi_1 = 1$$

Variables: a unit root implies random-walk-type non-stationarity.

ADF regression

$$\Delta y_t = c_t + \phi_c y_{t-1} + \beta_1 \Delta y_{t-1} + \dots + \beta_p \Delta y_{t-p} + \epsilon_t$$

Variables: $\Delta y_t = y_t - y_{t-1}$; c_t is deterministic component; ϕ_c is the unit-root test coefficient; β_j controls lagged differences.

ADF coefficient relationship

$$\phi_c = \phi_1 - 1$$

Variables: $\phi_c = 0$ corresponds to $\phi_1 = 1$.

ADF hypotheses

$$\begin{aligned} H_0 : \phi_c &= 0 && \text{unit root / non-stationary} \\ H_1 : \phi_c &< 0 && \text{stationary} \end{aligned}$$

Variables: the ADF alternative is one-sided and negative.

ADF statistic

$$\text{ADF} = \frac{\hat{\phi}_c}{\text{se}(\hat{\phi}_c)}$$

Variables: $\hat{\phi}_c$ is the estimated ADF coefficient.

ADF decision rule

$$\text{ADF} < \text{ADF critical value} \Rightarrow \text{reject } H_0$$

Variables: “less than” means more negative than the critical value.

Drift-only ADF deterministic component

$$c_t = c$$

Variables: c is a constant drift/intercept.

Trend ADF deterministic component

$$c_t = c_0 + c_1 t$$

Variables: c_0 is an intercept; c_1 is deterministic trend slope.

Martingale difference sequence

$$\mathbb{E}(\epsilon_t \mid F_{t-1}) = 0$$

Variables: ϵ_t has zero conditional mean given past information.

AR(1) model

$$y_t = c + \phi_1 y_{t-1} + \epsilon_t$$

Variables: c is intercept; ϕ_1 is the AR(1) coefficient.

AR(1) conditional mean

$$\mathbb{E}(y_t \mid F_{t-1}) = c + \phi_1 y_{t-1}$$

Variables: the conditional mean is the one-step forecast given past information.

AR(1) unconditional mean

$$\mathbb{E}(y_t) = \frac{c}{1 - \phi_1}$$

Variables: this requires stationarity, typically $|\phi_1| < 1$.

AR(1) conditional variance

$$\text{Var}(\epsilon_t \mid F_{t-1}) = \sigma_\epsilon^2$$

Variables: σ_ϵ^2 is innovation variance.

$$\text{Var}(y_t \mid F_{t-1}) = \sigma_\epsilon^2$$

Variables: uncertainty in y_t conditional on F_{t-1} comes from the new shock.

AR(1) unconditional variance

$$\text{Var}(y_t) = \frac{\sigma_\epsilon^2}{1 - \phi_1^2}$$

Variables: this requires stationarity.

AR(1) one-step forecast

$$\mathbb{E}(y_{t+1} \mid F_t) = c + \phi_1 y_t$$

Variables: y_t is known at time t .

AR(1) two-step forecast

$$\mathbb{E}(y_{t+2} \mid F_t) = c + \phi_1 \mathbb{E}(y_{t+1} \mid F_t) = c(1 + \phi_1) + \phi_1^2 y_t$$

Variables: the unknown future y_{t+1} is replaced by its forecast.

AR(1) long-horizon forecast

$$\mathbb{E}(y_{t+h} \mid F_t) \rightarrow \frac{c}{1 - \phi_1}$$

Variables: forecasts mean-revert to the unconditional mean when the AR(1) is stationary.

ARCH and Volatility Models

White noise

$$\begin{aligned}\mathbb{E}(u_t) &= 0 \\ \text{Var}(u_t) &= \sigma^2 \\ \text{Corr}(u_t, u_{t-k}) &= 0, \quad k \geq 1\end{aligned}$$

Variables: u_t is a white-noise shock.

MDS conditional mean

$$\mathbb{E}(u_t \mid F_{t-1}) = 0$$

Variables: an MDS is unpredictable from past information.

MDS implies zero unconditional mean

$$\mathbb{E}(u_t) = \mathbb{E}[\mathbb{E}(u_t \mid F_{t-1})] = 0$$

Variables: this uses the law of iterated expectations.

MDS implies no autocovariance

$$\text{Cov}(u_t, u_{t-k}) = 0, \quad k \geq 1$$

Variables: u_{t-k} is known at time $t-1$.

AR(p) model

$$y_t = c + \phi_1 y_{t-1} + \dots + \phi_p y_{t-p} + \epsilon_t$$

Variables: p is the autoregressive lag order.

AR(p) unconditional mean

$$\mathbb{E}(y_t) = \frac{c}{1 - \phi_1 - \dots - \phi_p}$$

Variables: denominator uses the sum of AR coefficients.

AR(p) one-step forecast

$$\mathbb{E}(y_{t+1} \mid F_t) = c + \phi_1 y_t + \phi_2 y_{t-1} + \dots + \phi_p y_{t-p+1}$$

Variables: all lagged y values on the right are known at time t .

AR(p) selection

$$\text{lowest AIC} = \text{preferred model}$$

Variables: AIC is Akaike information criterion.

Mean equation with residual

$$r_t = \mathbb{E}(r_t \mid F_{t-1}) + \epsilon_t$$

Variables: ϵ_t is the return shock after using information at $t-1$.

Constant-variance benchmark

$$\text{Var}(\epsilon_t \mid F_{t-1}) = \sigma_\epsilon^2$$

Variables: this benchmark has no time-varying conditional volatility.

ARCH LM mean model

$$y_t = x_t' \beta + \epsilon_t$$

Variables: x_t is a regressor vector; β is a coefficient vector.

ARCH LM auxiliary regression

$$\hat{\epsilon}_t^2 = \rho_0 + \rho_1 \hat{\epsilon}_{t-1}^2 + \dots + \rho_q \hat{\epsilon}_{t-q}^2 + v_t$$

Variables: $\hat{\epsilon}_t^2$ is squared residual; q is the number of ARCH lags tested.

ARCH LM hypotheses and statistic

$$\begin{aligned} H_0 : \rho_1 = \dots = \rho_q = 0 & \quad \text{no ARCH effects} \\ H_1 : \text{at least one } \rho_j \neq 0 & \quad \text{ARCH effects exist} \\ TR^2 & \sim \chi_q^2 \end{aligned}$$

Variables: R^2 comes from the auxiliary regression.

ARCH LM decision rule

$$TR^2 > \chi_q^2 \text{ critical value} \Rightarrow \text{reject } H_0$$

Variables: rejecting means volatility is time-varying and predictable.

ARCH model structure

$$\epsilon_t = \sigma_t u_t, \quad u_t \sim \text{iid}(0, 1)$$

Variables: σ_t is conditional standard deviation; u_t is a standardised shock.

ARCH conditional moments

$$\mathbb{E}(\epsilon_t \mid F_{t-1}) = 0, \quad \text{Var}(\epsilon_t \mid F_{t-1}) = \sigma_t^2$$

Variables: σ_t^2 is conditional variance.

ARCH(1)

$$\sigma_t^2 = \alpha_0 + \alpha_1 \epsilon_{t-1}^2$$

Variables: α_0 is variance intercept; α_1 is the ARCH coefficient.

ARCH(1) conditions

$$\alpha_0 > 0, \quad \alpha_1 \geq 0, \quad \alpha_1 < 1$$

Variables: these keep variance positive and finite.

ARCH(q)

$$\sigma_t^2 = \alpha_0 + \alpha_1 \epsilon_{t-1}^2 + \dots + \alpha_q \epsilon_{t-q}^2$$

Variables: q is the number of lagged squared shocks.

ARCH(q) conditions

$$\alpha_0 > 0, \quad \alpha_i \geq 0, \quad \sum_{i=1}^q \alpha_i < 1$$

Variables: $\sum \alpha_i$ measures volatility persistence in ARCH(q).

ARCH(1) unconditional moments

$$\begin{aligned} \mathbb{E}(\epsilon_t) &= 0 \\ \text{Cov}(\epsilon_t, \epsilon_{t-k}) &= 0, \quad k \geq 1 \\ \text{Var}(\epsilon_t) &= \frac{\alpha_0}{1 - \alpha_1} \end{aligned}$$

Variables: ARCH errors are uncorrelated, but squared errors can be autocorrelated.

Information criteria

$$AIC = -2\ell + 2k, \quad BIC = -2\ell + k \log(T)$$

Variables: ℓ is log-likelihood; k is number of estimated parameters; lower is better.

GARCH, Forecasting Volatility, and Model Comparison

ARCH(q) variance equation

$$\sigma_t^2 = \alpha_0 + \alpha_1 \epsilon_{t-1}^2 + \dots + \alpha_q \epsilon_{t-q}^2$$

Variables: this is the pure ARCH variance equation.

GARCH(p,q)

$$\begin{aligned} \epsilon_t &= \sigma_t u_t, \quad u_t \sim \text{iid}(0, 1) \\ \sigma_t^2 &= \alpha_0 + \alpha_1 \epsilon_{t-1}^2 + \dots + \alpha_q \epsilon_{t-q}^2 \\ &\quad + \beta_1 \sigma_{t-1}^2 + \dots + \beta_p \sigma_{t-p}^2 \end{aligned}$$

Variables: β_j are coefficients on lagged conditional variances; p is GARCH order; q is ARCH order.

GARCH(1,1)

$$\sigma_t^2 = \alpha_0 + \alpha_1 \epsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2$$

Variables: α_1 is the shock reaction; β_1 is volatility persistence from lagged variance.

GARCH(1,1) conditions

$$\alpha_0 > 0, \quad \alpha_1 \geq 0, \quad \beta_1 \geq 0, \quad \alpha_1 + \beta_1 < 1$$

Variables: $\alpha_1 + \beta_1 < 1$ gives finite unconditional variance and mean reversion.

GARCH persistence

$$\rho = \alpha_1 + \beta_1$$

Variables: ρ measures persistence of volatility shocks.

GARCH long-run variance and volatility

$$\text{Var}(\epsilon_t) = \bar{\sigma}^2 = \frac{\alpha_0}{1 - \alpha_1 - \beta_1}, \quad \bar{\sigma} = \sqrt{\frac{\alpha_0}{1 - \alpha_1 - \beta_1}}$$

Variables: $\bar{\sigma}^2$ is long-run variance; $\bar{\sigma}$ is long-run volatility.

GARCH conditional and unconditional moments

$$\begin{aligned}\mathbb{E}(\epsilon_t \mid F_{t-1}) &= 0 \\ \text{Var}(\epsilon_t \mid F_{t-1}) &= \sigma_t^2 \\ \text{Cov}(\epsilon_t, \epsilon_{t-k} \mid F_{t-1}) &= 0\end{aligned}$$

Variables: conditional variance changes over time even when conditional mean is zero.

$$\begin{aligned}\mathbb{E}(\epsilon_t) &= 0 \\ \text{Cov}(\epsilon_t, \epsilon_{t-k}) &= 0 \\ \text{Var}(\epsilon_t) &= \frac{\alpha_0}{1 - \alpha_1 - \beta_1}\end{aligned}$$

Variables: standard GARCH errors are uncorrelated but not independent in squares.

Normal GARCH innovation skewness

$$u_t \sim \text{iid}\mathcal{N}(0, 1), \quad \epsilon_t \mid F_{t-1} \sim \mathcal{N}(0, \sigma_t^2)$$

Variables: conditional normality makes conditional skewness zero.

$$\mathbb{E}(\epsilon_t^3 \mid F_{t-1}) = 0, \quad \mathbb{E}(\epsilon_t^3) = \mathbb{E}[\mathbb{E}(\epsilon_t^3 \mid F_{t-1})] = 0$$

Variables: the unconditional third moment is zero under symmetric normal innovations.

$$\text{skewness} = \frac{\mathbb{E}(\epsilon_t^3)}{[\text{Var}(\epsilon_t)]^{3/2}} = 0$$

Variables: skewness is zero for standard symmetric GARCH innovations.

Mean equation plus GARCH volatility equation

$$\begin{aligned}r_t &= c + \phi_1 r_{t-1} + \epsilon_t \\ \sigma_t^2 &= \alpha_0 + \alpha_1 \epsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2\end{aligned}$$

Variables: c, ϕ_1 define the mean equation; the second line defines volatility.

Constant-mean return equation

$$r_t = \mu + \epsilon_t$$

Variables: μ is constant mean return.

Mean-adjusted AR form

$$r_t - \mu = \phi(r_{t-1} - \mu) + \epsilon_t$$

Variables: ϕ is the AR coefficient around mean μ .

$$r_t = c + \phi r_{t-1} + \epsilon_t, \quad c = \mu(1 - \phi)$$

Variables: c is the equivalent intercept.

$$r_t = \mu + \phi(r_{t-1} - \mu) + \epsilon_t = \mu(1 - \phi) + \phi r_{t-1} + \epsilon_t$$

Variables: these are algebraically equivalent AR(1) mean forms.

AR(1)-GARCH(1,1) one-step mean forecast

$$\mathbb{E}(r_{T+1} \mid F_T) = \hat{r}_{T+1} = c + \phi r_T$$

Variables: r_T is the latest known return.

GARCH one-step variance forecast

$$\hat{\sigma}_{T+1}^2 = \mathbb{E}(\sigma_{T+1}^2 \mid F_T) = \alpha_0 + \alpha_1 \epsilon_T^2 + \beta_1 \sigma_T^2$$

Variables: ϵ_T^2 is the latest squared shock; σ_T^2 is the latest conditional variance.

Prediction interval using forecast variance

$$r_{T+1} \mid F_T \sim \mathcal{N}(\text{mean forecast, variance forecast})$$

Variables: mean forecast is $\hat{r}_{T+1}$; variance forecast is $\hat{\sigma}_{T+1}^2$.

$$\text{mean forecast} \pm 1.96 \sqrt{\text{variance forecast}}$$

Variables: 1.96 is the approximate 95 percent normal cutoff.

Likelihood-ratio comparison of ARCH and GARCH

$$LR = 2(LL_{GARCH} - LL_{ARCH}) = 2(\ell_1 - \ell_0)$$

Variables: LL_{GARCH} is the unrestricted log likelihood; LL_{ARCH} is the restricted log likelihood.

$$LR \sim \chi_p^2$$

Variables: p is the number of restrictions or extra parameters tested.

Model comparison rules

higher log likelihood $\Rightarrow$ better

lower AIC/BIC $\Rightarrow$ better

squared standardised residuals with no autocorrelation $\Rightarrow$ better diagnostics

Variables: these compare fitted volatility models.

Asymmetric Volatility Models

Standard GARCH symmetry

$$\sigma_t^2 = \alpha_0 + \alpha_1 \epsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2$$

Variables: squaring removes the sign of the shock.

$$(+a)^2 = (-a)^2 = a^2$$

Variables: a is any shock size.

GJR-GARCH model

$$r_t = \mu + \epsilon_t$$

$$\epsilon_t = \sigma_t u_t$$

$$\sigma_t^2 = \alpha_0 + \alpha_1 \epsilon_{t-1}^2 + \lambda I_{t-1} \epsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2$$

Variables: λ is the leverage/asymmetry coefficient; I_{t-1} is a bad-news indicator.

GJR indicator

$$I_{t-1} = \begin{cases} 1, & \epsilon_{t-1} \leq 0 \\ 0, & \text{otherwise} \end{cases}$$

Variables: the indicator switches on for non-positive shocks.

GJR good-news and bad-news effects

$$\begin{aligned} \epsilon_{t-1} > 0 : & \quad \text{effect} = \alpha_1 \epsilon_{t-1}^2 \\ \epsilon_{t-1} \leq 0 : & \quad \text{effect} = (\alpha_1 + \lambda) \epsilon_{t-1}^2 \end{aligned}$$

Variables: good news uses only α_1 ; bad news uses $\alpha_1 + \lambda$.

$$\text{good news effect} = \alpha_1, \quad \text{bad news effect} = \alpha_1 + \lambda$$

Variables: a positive λ means bad news has a larger variance effect in GJR.

GJR leverage test

$$\begin{aligned} H_0 : \lambda &= 0 \\ H_1 : \lambda &\neq 0 \\ t &= \frac{\hat{\lambda}}{\text{se}(\hat{\lambda})} \end{aligned}$$

Variables: $\hat{\lambda}$ is the estimated leverage coefficient.

$$|t| > 1.96 \Rightarrow \text{reject } H_0 \text{ at approximately 5 percent}$$

Variables: rejecting means statistically significant asymmetry.

GJR parameter conditions

$$\alpha_0 > 0, \quad \alpha_1 \geq 0, \quad \beta_1 \geq 0, \quad \alpha_1 + \lambda \geq 0$$

Variables: these keep conditional variance non-negative for good and bad news.

GJR unconditional variance

$$\text{Var}(\epsilon_t) = \frac{\alpha_0}{1 - \alpha_1 - \beta_1 - \lambda/2}$$

Variables: $\lambda/2$ appears when positive and negative shocks are equally likely.

EGARCH model

$$\log(\sigma_t^2) = \omega + \alpha_1 u_{t-1} + \gamma_1 (|u_{t-1}| - \mathbb{E}|u_{t-1}|) + \beta_1 \log(\sigma_{t-1}^2)$$

Variables: ω is intercept; α_1 captures signed shock asymmetry; γ_1 captures magnitude effects; β_1 captures log-volatility persistence.

EGARCH standardised shock

$$u_{t-1} = \frac{\epsilon_{t-1}}{\sigma_{t-1}}$$

Variables: u_{t-1} standardises the shock by conditional volatility.

Expected absolute standard normal shock

$$\mathbb{E}|u_t| = \sqrt{\frac{2}{\pi}} \approx 0.7979$$

Variables: this value is used when u_t is standard normal.

EGARCH positivity

$$\log(\sigma_t^2) \text{ is modelled } \Rightarrow \sigma_t^2 \text{ is automatically positive}$$

Variables: modelling log variance avoids non-negativity constraints on coefficients.

EGARCH leverage sign in the notes

$$\lambda < 0 \Rightarrow \text{negative shocks raise volatility more than positive shocks}$$

Variables: λ is the asymmetry coefficient in the note's EGARCH interpretation.

News impact curve for ARCH(1)

$$NIC(\epsilon_{t-1}) = \sigma_t^2 = \alpha_0 + \alpha_1 \epsilon_{t-1}^2$$

Variables: NIC maps yesterday's shock into today's conditional variance.

News impact curve for GARCH(1,1)

$$\bar{\sigma}^2 = \frac{\alpha_0}{1 - \alpha_1 - \beta_1}$$

Variables: $\bar{\sigma}^2$ is the variance level used to hold lagged variance fixed.

$$NIC(\epsilon_{t-1}) = \alpha_0 + \alpha_1 \epsilon_{t-1}^2 + \beta_1 \bar{\sigma}^2$$

Variables: the curve varies with ϵ_{t-1} while holding $\sigma_{t-1}^2 = \bar{\sigma}^2$.

$$NIC(\epsilon_{t-1}) = \alpha_0 + \beta_1 \frac{\alpha_0}{1 - \alpha_1 - \beta_1} + \alpha_1 \epsilon_{t-1}^2$$

Variables: this substitutes the long-run variance into the NIC.

News impact curve for GJR-GARCH

$$A = \alpha_0 + \beta_1 \frac{\alpha_0}{1 - \alpha_1 - \lambda/2 - \beta_1}$$

Variables: A is the intercept-like NIC component after fixing lagged variance at the GJR long-run variance.

$$\begin{aligned} \epsilon_{t-1} > 0 : \quad \sigma_t^2 &= A + \alpha_1 \epsilon_{t-1}^2 \\ \epsilon_{t-1} \leq 0 : \quad \sigma_t^2 &= A + (\alpha_1 + \lambda) \epsilon_{t-1}^2 \end{aligned}$$

Variables: the two branches show asymmetric curvature for good and bad news.

IGARCH, RiskMetrics, MA, and ARMA

IGARCH persistence

$$\alpha_1 + \beta_1 = 1$$

Variables: IGARCH has unit volatility persistence.

IGARCH(1,1) from GARCH(1,1)

$$\sigma_t^2 = \alpha_0 + \alpha_1 \epsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2, \quad \alpha_1 = 1 - \beta_1$$

Variables: the second condition enforces $\alpha_1 + \beta_1 = 1$.

IGARCH with zero intercept

$$\alpha_0 = 0$$

Variables: RiskMetrics/EWMA uses zero variance intercept.

$$\sigma_t^2 = (1 - \beta_1) \epsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2$$

Variables: β_1 is the decay weight on old variance.

IGARCH using returns as shocks

$$\sigma_t^2 = (1 - \beta_1) r_{t-1}^2 + \beta_1 \sigma_{t-1}^2$$

Variables: this uses r_{t-1} as the shock when the conditional mean is zero.

IGARCH unconditional variance issue

$$\text{Var}(\epsilon_t) = \frac{\alpha_0}{1 - \alpha_1 - \beta_1}, \quad \alpha_1 + \beta_1 = 1$$

Variables: the denominator is zero, so stationary unconditional variance is not finite.

RiskMetrics conditional normal model

$$r_t \mid F_{t-1} \sim \mathcal{N}(0, \sigma_t^2)$$

Variables: RiskMetrics often assumes zero conditional mean.

RiskMetrics variance update

$$\sigma_t^2 = \lambda \sigma_{t-1}^2 + (1 - \lambda) r_{t-1}^2$$

Variables: λ is the decay factor; $1 - \lambda$ is the latest squared-return weight.

RiskMetrics one-step forecast

$$\sigma_{T+1}^2 = \lambda \sigma_T^2 + (1 - \lambda) r_T^2$$

Variables: r_T is the latest observed return.

Conditional-normal VaR

$$r_{T+1} \mid F_T \sim \mathcal{N}(\mu_{T+1|T}, \sigma_{T+1}^2)$$

Variables: $\mu_{T+1|T}$ is the conditional mean forecast.

$$q_\alpha = \mu_{T+1|T} + \sigma_{T+1} z_\alpha$$

Variables: q_α is the conditional return quantile.

$$\text{VaR}_\alpha = |W_0 q_\alpha|$$

Variables: W_0 is dollar exposure.

$$q_{0.05} = -1.645\sigma_{T+1}, \quad \text{VaR}_{0.05} = |W_0(-1.645\sigma_{T+1})|$$

Variables: this zero-mean form uses $z_{0.05} = -1.645$.

AR-GARCH VaR model

$$\begin{aligned} r_t &= \mu_t + \epsilon_t \\ \mu_t &= c + \phi_1 r_{t-1} + \dots + \phi_p r_{t-p} \\ \sigma_t^2 &= \alpha_0 + \alpha_1 \epsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2 \end{aligned}$$

Variables: μ_t is conditional mean; p is AR order.

AR-GARCH mean forecast

$$\mu_{T+1|T} = c + \phi_1 r_T + \dots + \phi_p r_{T-p+1}$$

Variables: all returns on the right are known at time T .

AR-GARCH VaR quantile

$$q_{0.05} = \mu_{T+1|T} - 1.645\sigma_{T+1}, \quad \text{VaR}_{0.05} = |W_0 q_{0.05}|$$

Variables: $\sigma_{T+1} = \sqrt{\sigma_{T+1}^2}$.

Persistence versus volatility level

$$\text{persistence} = \alpha_1 + \beta_1, \quad \text{long-run variance} = \frac{\alpha_0}{1 - \alpha_1 - \beta_1}$$

Variables: persistence controls shock decay; α_0 also affects the volatility level.

MA(1) model

$$y_t = \phi_0 + u_t + \theta_1 u_{t-1}$$

Variables: ϕ_0 is mean/intercept; θ_1 is MA coefficient; u_t is shock.

$$u_t \sim \text{iid}(0, \sigma_u^2)$$

Variables: σ_u^2 is innovation variance.

MA(1) mean, variance, and autocorrelation

$$\mathbb{E}(y_t) = \phi_0, \quad \text{Var}(y_t) = (1 + \theta_1^2)\sigma_u^2$$

Variables: MA(1) variance depends on θ_1^2 .

$$\rho_1 = \frac{\theta_1}{1 + \theta_1^2}, \quad \rho_k = 0 \text{ for } k > 1$$

Variables: MA(1) autocorrelation cuts off after lag 1.

MA(1) forecasts

$$y_{T+1} = \phi_0 + u_{T+1} + \theta_1 u_T$$

Variables: u_T is the latest known/estimated shock.

$$\hat{y}_{T+1} = \hat{\phi}_0 + \hat{\theta}_1 \hat{u}_T, \quad \hat{y}_{T+h} = \hat{\phi}_0 \text{ for } h \geq 2$$

Variables: future shocks have zero expected value.

ARMA(p,q)

$$y_t = \phi_0 + \phi_1 y_{t-1} + \dots + \phi_p y_{t-p} + u_t + \theta_1 u_{t-1} + \dots + \theta_q u_{t-q}$$

Variables: p is AR order; q is MA order.

lowest AIC = preferred

Variables: use AIC to choose among candidate ARMA models.

Multi-Step Forecasting and Portfolio VaR

Future squared shocks equal future variance in expectation

$$\mathbb{E}(\epsilon_{t+h-1}^2 \mid F_t) = \mathbb{E}(\sigma_{t+h-1}^2 \mid F_t)$$

Variables: future squared shocks are unknown, so use their conditional expectation.

$$\epsilon_{t+h-1} = \sigma_{t+h-1} u_{t+h-1}, \quad \mathbb{E}(u_{t+h-1}^2 \mid F_{t+h-2}) = 1$$

Variables: u is standardised to have conditional second moment 1.

Recursive h-step GARCH variance forecast

$$\rho = \alpha_1 + \beta_1$$

Variables: ρ is GARCH persistence.

$$\mathbb{E}(\sigma_{t+h}^2 \mid F_t) = \alpha_0 + \rho \mathbb{E}(\sigma_{t+h-1}^2 \mid F_t)$$

Variables: the h -step forecast uses the previous horizon's variance forecast.

Closed-form h-step GARCH forecast

$$\begin{aligned} \mathbb{E}(\sigma_{t+h}^2 \mid F_t) &= \alpha_0 \frac{1 - \rho^{h-1}}{1 - \rho} \\ &\quad + \rho^{h-1} \mathbb{E}(\sigma_{t+1}^2 \mid F_t) \end{aligned}$$

Variables: h is forecast horizon; ρ^{h-1} controls mean reversion speed.

Closed-form forecast using long-run variance

$$\bar{\sigma}^2 = \frac{\alpha_0}{1 - \alpha_1 - \beta_1}$$

Variables: $\bar{\sigma}^2$ is long-run variance.

$$\mathbb{E}(\sigma_{t+h}^2 \mid F_t) = \bar{\sigma}^2 + \rho^{h-1} [\mathbb{E}(\sigma_{t+1}^2 \mid F_t) - \bar{\sigma}^2]$$

Variables: forecasts converge to $\bar{\sigma}^2$ when $\rho < 1$.

$$\mathbb{E}(\sigma_{t+h}^2 \mid F_t) \rightarrow \bar{\sigma}^2$$

Variables: this is the long-horizon mean-reversion result.

Prediction interval for returns

$$r_{T+1} \mid F_T \sim \mathcal{N}(\mu_{T+1|T}, \sigma_{T+1}^2)$$

Variables: $\mu_{T+1|T}$ is the one-step mean forecast; σ_{T+1}^2 is the one-step variance forecast.

$$\mu_{T+1|T} \pm 1.96\sqrt{\sigma_{T+1}^2}$$

Variables: this is an approximate 95 percent prediction interval.

One-asset VaR with forecast variance

$$q_{0.05} = \mu_{T+1|T} - 1.645\sqrt{\sigma_{T+1}^2}$$

Variables: $q_{0.05}$ is the 5 percent bad-return quantile.

$$\text{VaR}_{0.05} = |W_0 q_{0.05}|$$

Variables: W_0 is position value.

$$\text{VaR}_{0.05} = |W_0(-1.645\sqrt{\sigma_{T+1}^2})|$$

Variables: this is the zero-mean special case.

Asset-specific RiskMetrics forecast

$$\sigma_{i,t+1}^2 = (1 - \lambda_i)r_{i,t}^2 + \lambda_i\sigma_{i,t}^2$$

Variables: i indexes the asset; λ_i is asset-specific decay factor.

Asset-specific RiskMetrics VaR

$$\text{VaR}_i = \left| W_i(-1.645)\sqrt{\sigma_{i,t+1}^2} \right|$$

Variables: W_i is the dollar exposure to asset i .

Two-asset portfolio return for VaR

$$r_{p,t} = wr_{m,t} + (1 - w)r_{c,t}$$

Variables: w is the portfolio weight in asset m ; $1 - w$ is the weight in asset c .

$$w = \frac{\text{amount in asset } m}{\text{total portfolio value}}, \quad 1 - w = \frac{\text{amount in asset } c}{\text{total portfolio value}}$$

Variables: weights are based on dollar values.

Two-asset portfolio variance for VaR

$$\sigma_p^2 = w^2\sigma_m^2 + (1 - w)^2\sigma_c^2 + 2w(1 - w)\rho\sigma_m\sigma_c$$

Variables: ρ is correlation between the two asset returns.

Portfolio VaR from portfolio volatility

$$\text{VaR}_p = |\text{total value} \times (-1.645) \times \sigma_p|$$

Variables: $\sigma_p = \sqrt{\sigma_p^2}$ is portfolio standard deviation.

Portfolio VaR from individual VaRs

$$\text{VaR}_p = \sqrt{\text{VaR}_m^2 + \text{VaR}_c^2 + 2\rho \text{VaR}_m \text{VaR}_c}$$

Variables: $\text{VaR}_m, \text{VaR}_c$ are stand-alone asset VaRs.

Generic two-asset portfolio variance and VaR notation

$$\sigma_p^2 = w_A^2\sigma_A^2 + w_B^2\sigma_B^2 + 2w_Aw_B\rho_{AB}\sigma_A\sigma_B$$

Variables: A, B are generic assets.

$$\text{VaR}_p = W_0(1.645)\sigma_p$$

Variables: this writes VaR as a positive loss number when mean return is zero.

$$\text{VaR}_p = \sqrt{\text{VaR}_A^2 + \text{VaR}_B^2 + 2\rho_{AB} \text{VaR}_A \text{VaR}_B}$$

Variables: this is the same normal portfolio VaR logic expressed using individual VaRs.

Essential Exam Answer Templates

ADF template

H_0 : unit root / non-stationary

H_1 : stationary

$$\text{ADF} = \frac{\text{coefficient}}{\text{standard error}}$$

reject only if ADF is more negative than the critical value

Variables: the coefficient is the ADF coefficient on lagged level.

General t-test template

$$\begin{aligned}H_0 &: \beta = \beta_0 \\H_1 &: \beta \neq \beta_0 \\t &= \frac{\hat{\beta} - \beta_0}{\text{se}(\hat{\beta})}\end{aligned}$$

compare $|t|$ to the critical value

Variables: β_0 is the hypothesised value.

CAPM versus multi-factor template

$$\begin{aligned}H_0 &: \text{extra factor coefficients are all zero} \\H_1 &: \text{at least one is non-zero} \\J &= \frac{RSS_0 - RSS_1}{RSS_1 / (T - K - 1)}\end{aligned}$$

compare J to chi-square with number-of-restrictions degrees of freedom

Variables: RSS_0 is restricted-model RSS; RSS_1 is unrestricted-model RSS.

ARCH test template

estimate mean model
square residuals
regress squared residuals on q lags
 $TR^2 \sim \chi_q^2$
reject means ARCH effects / time-varying volatility

Variables: q is number of lagged squared residuals in the auxiliary regression.

One-step GARCH forecast template

mean forecast = fitted mean equation using known latest returns
variance forecast = $\alpha_0 + \alpha_1 \epsilon_T^2 + \beta_1 \sigma_T^2$

Variables: the latest return, shock, and variance are known at time T .

Prediction interval template

$$\text{mean forecast} \pm 1.96 \sqrt{\text{variance forecast}}$$

Variables: use 1.96 for an approximate 95 percent interval under normality.

5 percent VaR template

$$\begin{aligned}q_{0.05} &= \text{mean forecast} - 1.645 \sqrt{\text{variance forecast}} \\ \text{VaR}_{0.05} &= |\text{position value} \times q_{0.05}|\end{aligned}$$

Variables: 1.645 is the one-sided 5 percent normal cutoff.